

Manuel Comi Xolot

Senior Frontend Engineer | React, TypeScript & High-Traffic Frontend Architecture

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SUMMARY

Senior Frontend Engineer with extensive experience building scalable, high-traffic web applications with React, TypeScript, and modern frontend architectures. At Harvard Business Publishing, I lead frontend modernization across a distributed microfrontend ecosystem serving millions of users, focusing on performance, maintainability, accessibility, and developer experience. My recent work includes AI-powered product features, accessibility-focused UI development, and state management patterns across microfrontend and monorepo environments. Recognized as a technical leader with impact beyond my immediate team: I drive architecture discussions, mentor engineers, improve CI/CD workflows, and partner with product, design, backend, and DevOps teams to deliver reliable user-facing experiences end-to-end.

EXPERIENCE

Unosquare — Senior Software Engineer, Team Lead 2021 – Present

Client: Harvard Business Publishing (Harvard Business Review & Harvard Business for Educators)

End-to-end frontend ownership across a distributed microfrontend platform for one of the world's leading business content brands, serving millions of monthly users.

Architecture, Platform Engineering & Technical Leadership

- Led frontend architecture decisions across concurrent initiatives, balancing long-term maintainability with fast product delivery.
- Helped modernize legacy monolithic experiences into a distributed microfrontend ecosystem, enabling more scalable, maintainable, and independently deployable frontend applications.
- Architected and evolved mfe-core, the shared component library powering the MFE ecosystem, including Storybook documentation, React 19 ecosystem upgrades, semantic-release versioning, and cross-team dependency upgrades.
- Partnered with product, design, backend, and DevOps teams to translate broad requirements into scalable frontend plans, reusable UI patterns, API contracts, and reliable release processes.
- Facilitated architecture discussions, reviewed peer code, mentored engineers, and raised standards for code quality, accessibility, testing, and maintainability.

Performance, Quality & Delivery

- Improved Core Web Vitals (LCP, FID) through SSR strategies, AVIF image optimization, CloudFront CDN caching, and frontend performance analysis on high-traffic editorial experiences.
- Implemented WCAG 2.1 accessibility standards and automated a11y and visual regression testing with Axe DevTools and Chromatic.
- Designed and maintained Jenkins CI/CD pipelines with Docker for QA, Ephemeral, and Production deployments, including health checks, rollback automation, deployment validation, and S3 sync operations.
- Owned complex frontend features from technical design through production deployment, unblocking teams and maintaining momentum in fast-paced product cycles.
- Contributed to AI-driven product features for HBP AI Advisor, including conversational interfaces, intent-based recommendations, asynchronous UI states, and contextual guidance integrated into existing product workflows.

AI Advisor — Harvard Business for Educators (Conversational AI Product)

Brought onto a separate HBP product (educator-facing SPA) to reinforce the team and ship a priority AI feature on schedule.

- Built core interface components for an AI-driven conversational advisor that helps educators discover cases, simulations, and course materials through natural language — from the chat experience to contextual recommendation cards.
- Re-architected the feature's state management for scalability, eliminating prop-drilling and centralizing data flow — and extracted reusable components into the shared component library for use across teams.

- Owned accessibility end-to-end (WCAG 2.1), resolving complex responsive layout defects and adding full screen-reader support and automated accessibility test coverage across the feature.

Coordinación de Nuevas Tecnologías e Informática — Filmoteca UNAM 2008 – 2021

Software Engineering Lead (2014–2021) · Senior Software Engineer (2012–2014) · Software Engineer (2008–2012)

Joined as the technical area was being established and grew into its engineering lead — building software and infrastructure from the ground up to manage and preserve Mexico's national film archive.

- Led end-to-end development of CLAF, a high-impact film collection management system — recognized by UNAM as an award-winning institutional engineering project — among other internal systems built to support digitization, preservation, and archive operations.
- Designed and delivered software where no prior infrastructure existed, establishing engineering practices, standards, and development workflows for the area.
- Mentored junior developers across full-stack projects (JavaScript, Java/Spring, Python, React, Node.js, MySQL, REST APIs), fostering their growth into professional roles.

SKILLS

Frontend: React, TypeScript/JavaScript, ES6, Next.js, HTML, CSS, Storybook, Zustand, CSS Modules, SASS, Responsive Design, Accessibility (WCAG 2.1), Performance Optimization

Architecture: Microfrontends, Design Systems, Component Libraries, Monorepos (Lerna), Semantic Versioning, Module Federation, State management (Context + reducers, Zustand), API-driven frontend architecture

Backend & APIs: Node.js, REST APIs, Java (Spring), microservices patterns, data modeling, API contracts, GraphQL (working knowledge)

Cloud & DevOps: AWS (S3, CloudFront, Lambda, Cognito, Secrets Manager), Docker, Jenkins, CI/CD pipelines

Testing & Quality: Jest, React Testing Library, jest-axe, Chromatic (visual regression), Axe DevTools, CI-integrated QA workflows, code reviews

AI & Tooling: Conversational UI, AI-powered product integrations, Copilot, Claude, ChatGPT, Codex, LaunchDarkly

EDUCATION

B.A. in Computer Science — Universidad del Valle de México (2013–2015)